

Italian_exam_A002_ORD23-English

Italian Maturità – Esame di Stato, Mathematics (A002)

Ordinary session 2023 – Second written test Source: *Ministero dell'istruzione e del merito*

Valid for the Scientific Lyceum (indirizzi LI02, LI03, LI15, LI1S, LI22, LI23, LI31, LI32, LIA2, LIAO, LIB2, LIC2, LID2, LII2, LII3, LII4, LIIS, LIS2, EA02, EA10).

Instructions: The candidate must solve **one** of the two problems and answer **4** of the questions in the questionnaire.

Problem 1

The graph in the figure represents the continuous function ($y=f(x)$) as the union of a parabolic arc ($_1$), a circular arc ($_2$), and a hyperbolic arc ($_3$).

(a) Write an analytic expression for the piecewise function (f) on the interval $([-2;2])$, using the equations $[y=a(x+2)^2, x^2+y^2+b=0, x^2-y^2+c=0,]$ and find the appropriate values of the real parameters (a, b, c). Study the differentiability of (f) and write the equations of the tangent lines (if any) at the points of abscissa ($x=-2, x=0, x=1, x=2$).

(b) Starting from the graph of (f), deduce the graph of its derivative (f') and identify the intervals of concavity and convexity of $[F(x)=-2^x \int_0^x f(t)dt.]$

(c) Consider the function ($y=(x+2)^2$), defined on the interval $([-2;0])$, of which ($_1$) is the graph. Explain why it is invertible and write the analytic expression of its inverse (h). Study the differentiability of (h) and sketch its graph.

(d) Let (S) be the bounded region of the second quadrant lying between the graph ($_1$) and the coordinate axes. Find the value of the real parameter (k) such that the line ($x=k$) divides (S) into two regions of equal area.

Problem 2

For a fixed real parameter (a), consider the function $[f_a(x)=,]$ whose graph is denoted ($_a$).

(a) As (a) varies, determine the domain of (f_a), study any discontinuities, and write the equations of all asymptotes.

(b) Show that, for (a), all graphs ($_a$) intersect their own horizontal asymptote at the same point, and share the same tangent line at the origin.

(c) For ($a<1$), find the intervals of monotonicity of (f_a). Study the function ($f_{-1}(x)$) and sketch its graph ($_{-1}$).

(d) Find the area of the bounded region between ($_{-1}$), its tangent line at the origin, and the line ($x=$).

Questionnaire (Quesiti)

1. Let (ABC) be a right triangle with the right angle at (A). Let (O) be the centre of the square (BCDE) constructed on the hypotenuse, on the opposite side from vertex (A). Prove that (O) is equidistant from the lines (AB) and (AC).
2. A loaded die, with faces numbered 1 to 6, has the property that each even face appears with twice the probability of each odd face. Calculate, for a single roll, the probabilities of obtaining:
 - a prime number;
 - a number at least equal to 3;
 - a number at most equal to 3.
3. Consider the line (r) through the points (A(1,-2,0)) and (B(2,3,-1)). Determine the Cartesian equation of the sphere with centre (C(1,-6,7)) tangent to (r).
4. Among all rectangular parallelepipeds with a square base and volume (V), determine whether the one with minimum total surface area also has the minimum-length diagonal.
5. Determine the equation of the tangent line to the curve (y=) at the point of abscissa 3, using two different methods.
6. Determine the values of the real parameters (a) and (b) such that $\int_{-\infty}^{+\infty} f(x) dx = 1$.
7. Consider the function $f(x) =$

$$\begin{cases} -1 + \arctan x, & x < 0 \\ ax + b, & x \geq 0 \end{cases}$$

-] Determine the values of the real parameters (a, b) for which the function is differentiable. State whether there exists an interval of $\mathbb{R}$ on which (f) satisfies the hypotheses of Rolle's theorem. Justify the answer.
8. Given the function $f_a(x) = x^5 - 5ax + a$, defined on all of $\mathbb{R}$, determine for which values of the parameter ($a > 0$) the function has three distinct real zeros.
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Maximum duration: 6 hours. Scientific or graphing calculators are allowed, provided they have no symbolic computation capability.